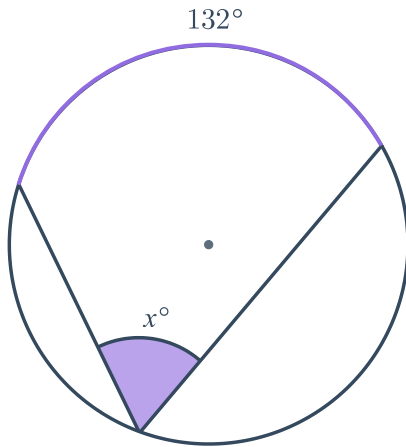


Inscribed angles and polygons

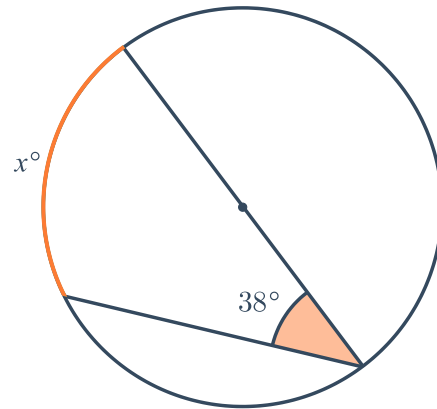


1 Solve for value of the variable(s) in the following diagrams.

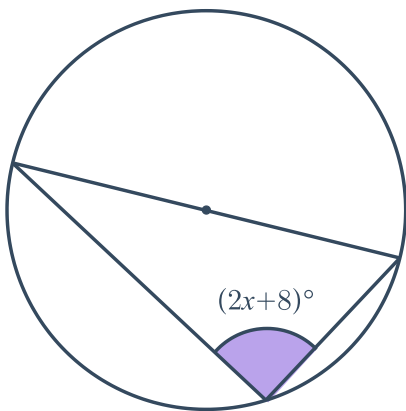
a



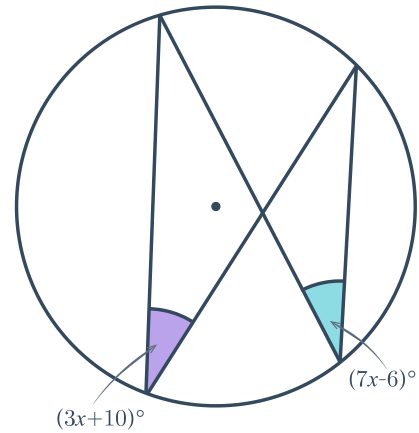
b



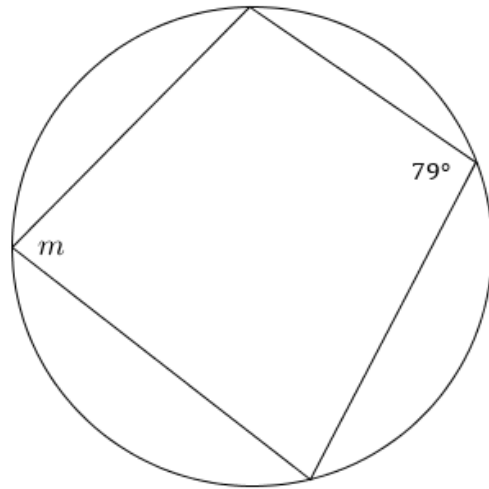
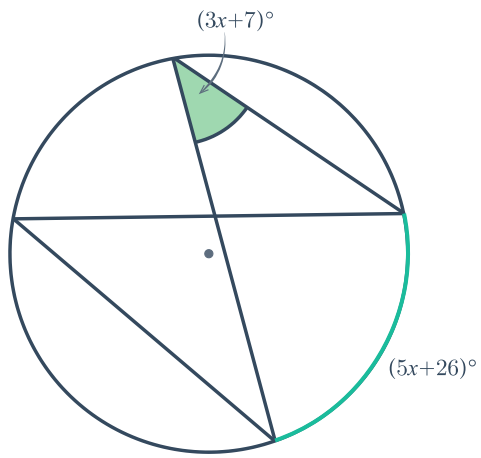
c



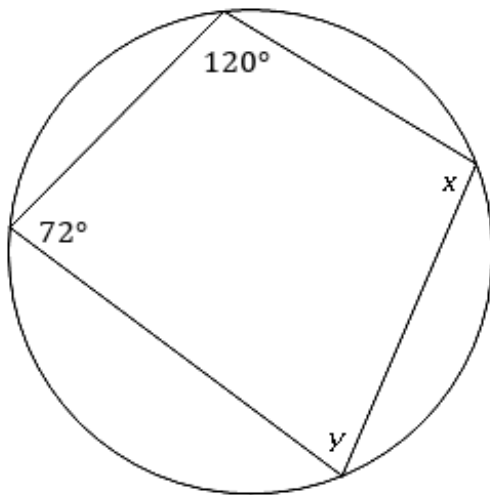
d



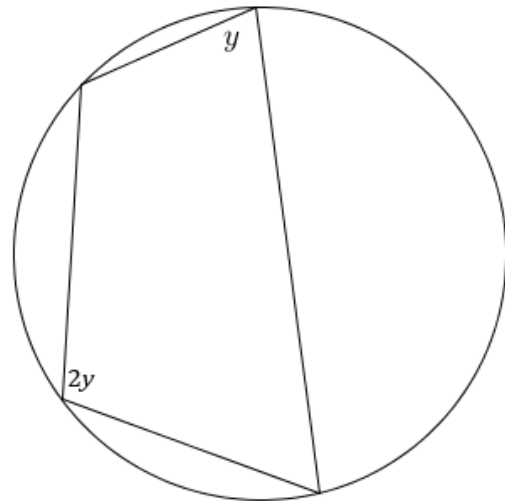
e



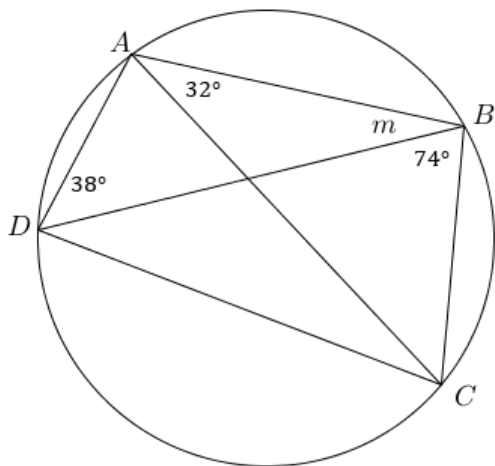
g



h



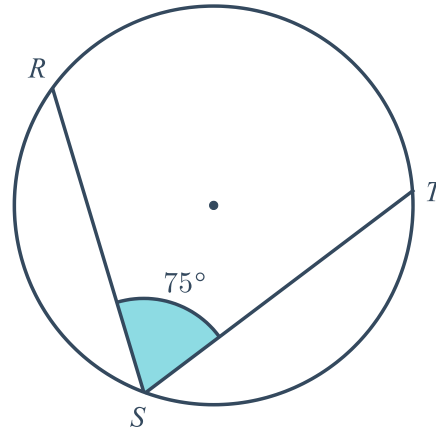
i



2 Consider the following diagram.

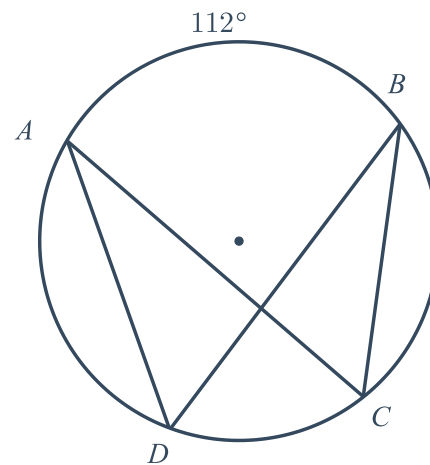


Find the measure of the minor arc $\widehat{RT}$.



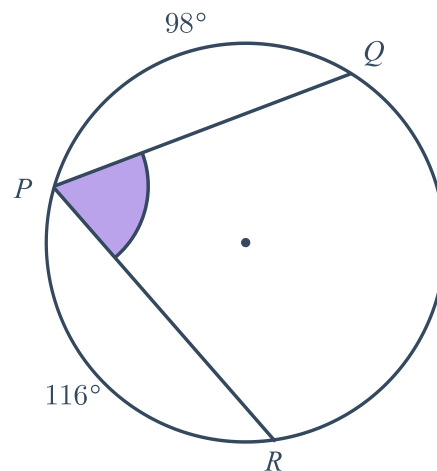
3 Consider the following diagram.

Find $m\angle ADB$.

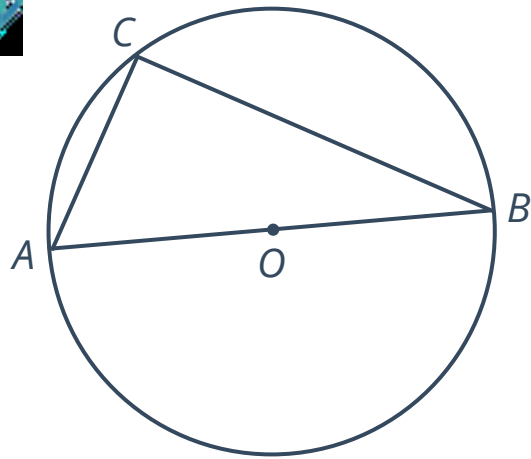


4 Consider the following diagram.

Find $m\angle RPQ$.

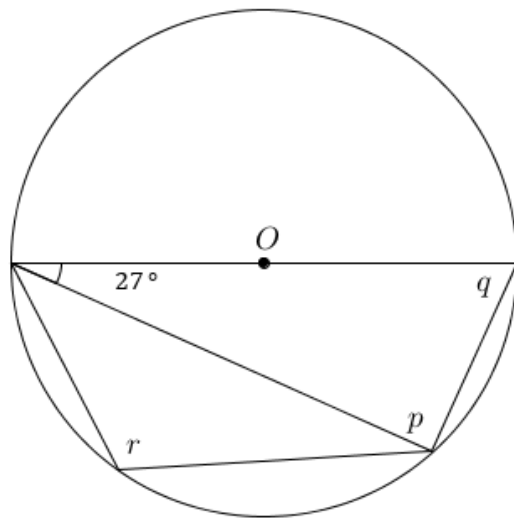


- 5 Consider the following diagram where A, B and O are collinear.
Find $m\angle ACB$.



- 6 In the diagram, O is the center of the circle.
A quadrilateral inscribed in the circle and one of its diagonal are drawn.

- a Solve for p .
- b Solve for q .
- c Solve for r .

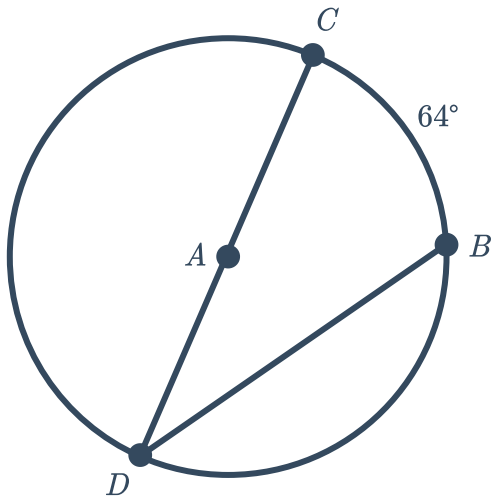




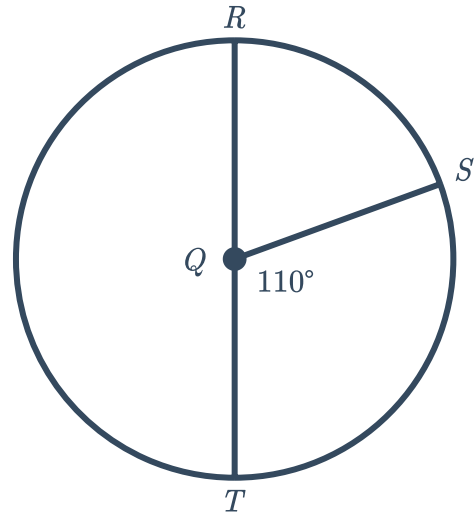
Additional questions

7

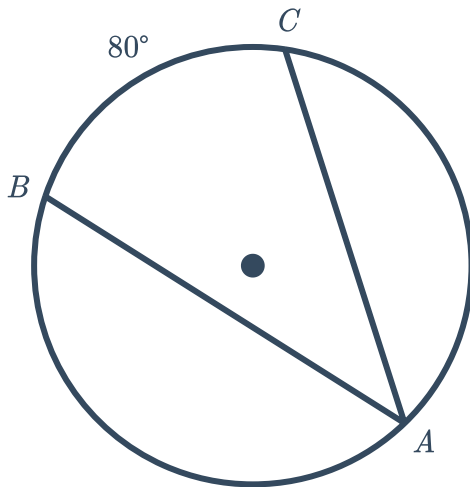
a Solve for $m\angle CDB$.



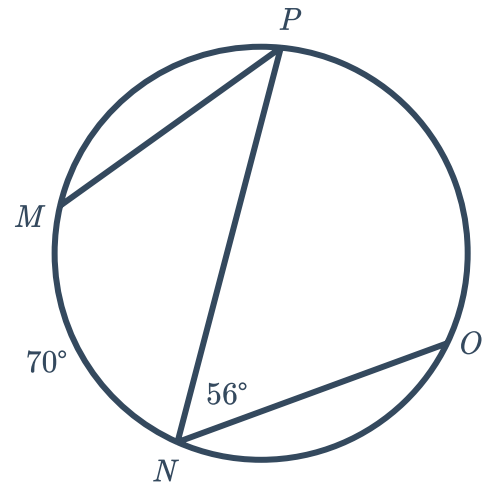
b Solve for $m\widehat{RS}$.



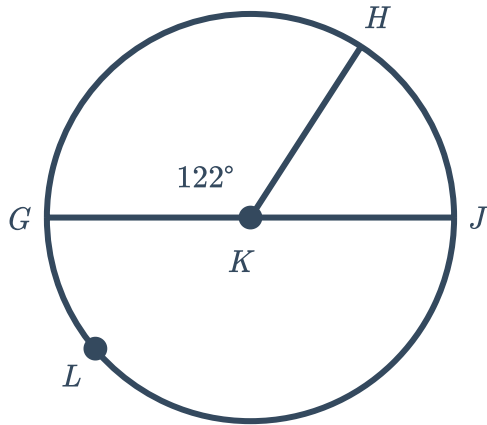
c Solve for $m\angle BAC$.



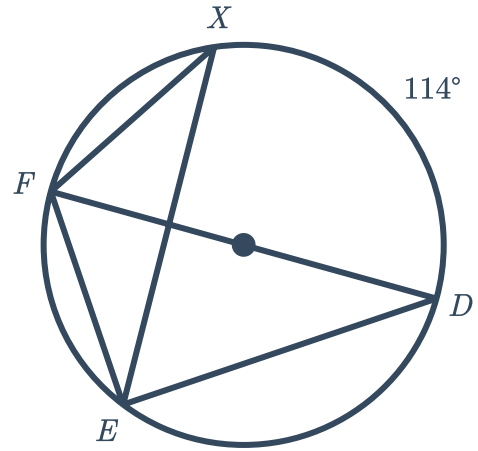
d Solve for $\angle MPN$.



e Solve for $m\widehat{GH}$.

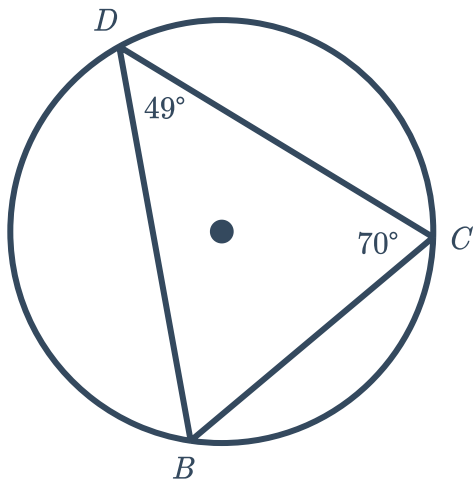


Solve for $m\angle XFD$.

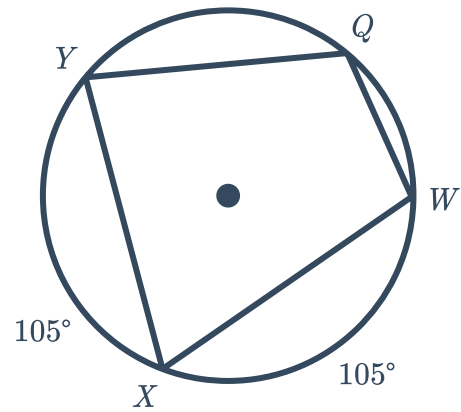


8

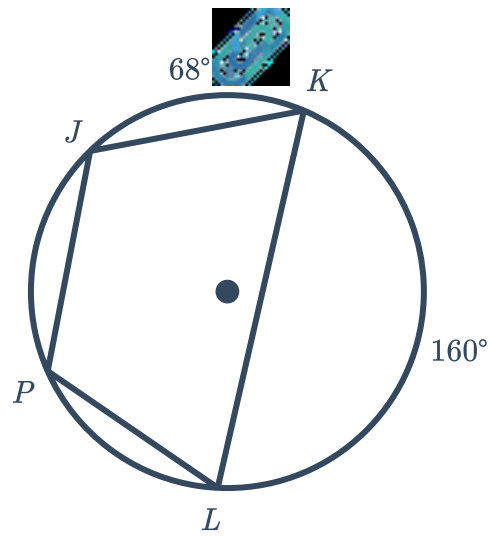
a Solve for $m\widehat{DC}$.



b Solve for $m\angle Q$.

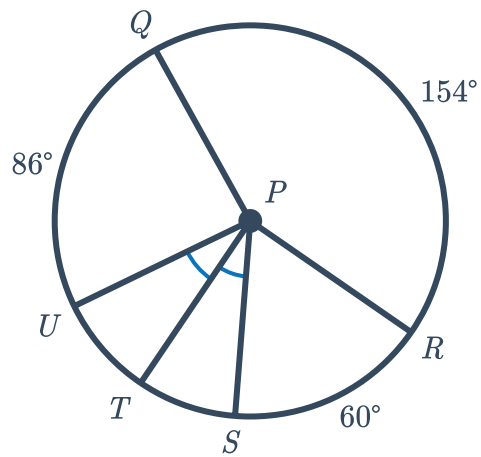


9



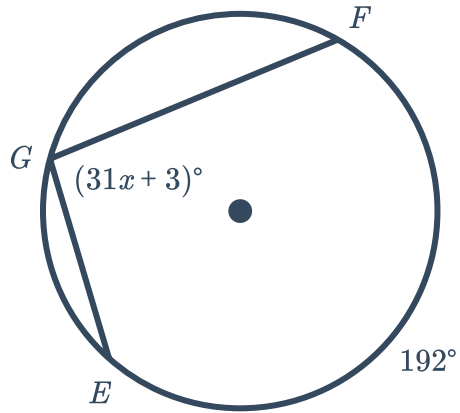
- a Solve for $m\angle P$
- b Solve for $m\angle K$

10

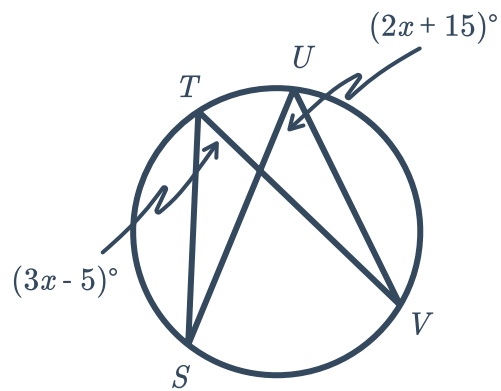


- a Solve for $m\angle UPT$
- b Solve for $m\widehat{US}$

11 Solve for x

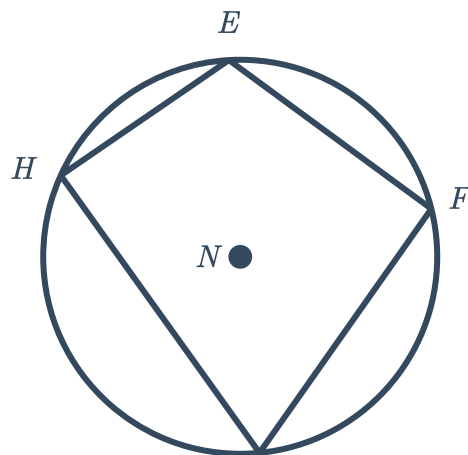


12



- a Solve for x
- b Solve for $m\widehat{SV}$

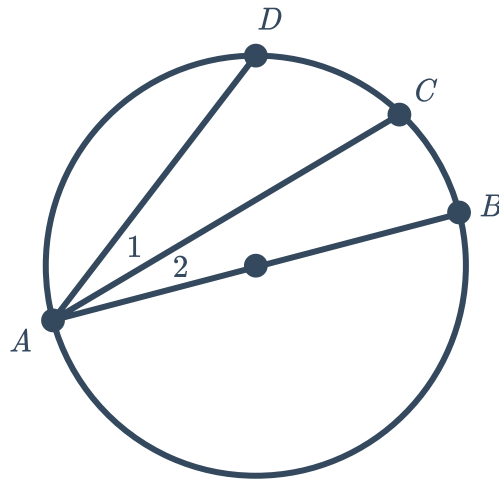
13 If $m\angle H = a^\circ$, what is $m\angle F$?





14 Given:

- The ratio of $m\widehat{BC} : m\widehat{BD}$ is 2 : 5.
- $m\angle 2 = 24^\circ$

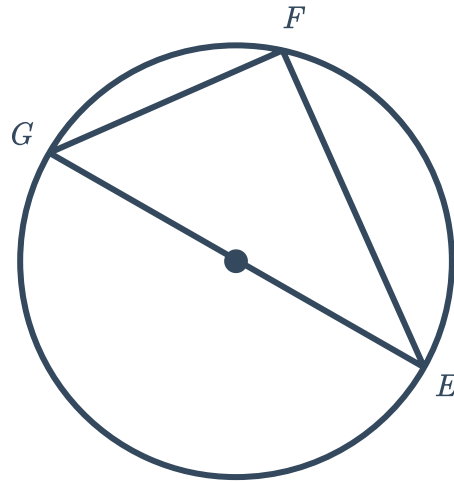


- a Solve for $m\angle 1$
- b Solve for $m\widehat{BD}$

15 Given:



- $\overline{GE}$ is a diameter of the circle
- $m\widehat{FE} = (9x + 18)^\circ$
- $m\angle G = 34^\circ$

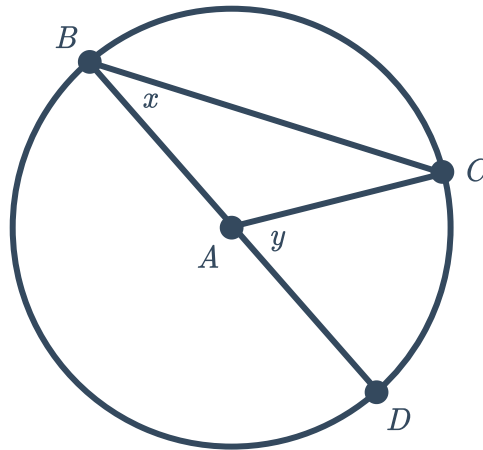


- a Solve for x
- b Solve for $m\widehat{GF}$

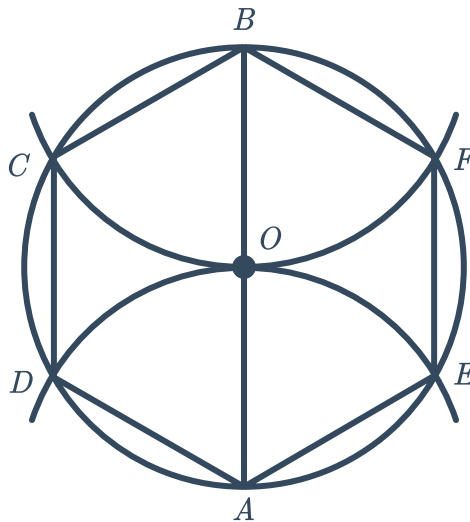
16 What is the relationship between the inscribed angles in the circle? Justify your answer.



17 Show that $y = 2x$.

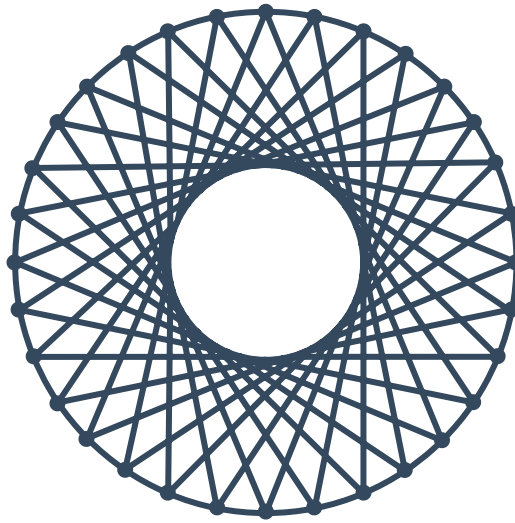


18 Andie constructed a regular hexagon inscribed in circle O :



- a What is the measure of each angle formed by two sides of the hexagon?
- b List all of the arcs with the same measure as $\widehat{CDA}$.
- c What is the relationship between $m\widehat{COF}$ and $m\widehat{CAF}$?

- 19 String art can be made by weaving string through various geometric patterns. This particular design consists of 32 equally spaced points, where the string has been woven to form angles that reach the points 12 positions away from its vertex on either side.



- a What is the measure of an angle formed by two segments sharing a common endpoint on the circle?
- b If the angles were modified so that each endpoint is 8 positions from the vertex of its angle, what would the measure of each be?
- c If the angles were modified so that each endpoint is n positions from the vertex of its angle, what would the measure of each be?